Animal Classification

Animal Classification: The formation of groups and sub-groups of animals depending upon the similarities and differences among them.

History of Animal Classification

- **7 Million Species:** According to recent studies, the estimated number of animal species on Earth is approximately 7 million, making it impossible to study each one individually without classification.
- **First Classification:** Greek philosopher **Aristotle** was the first to classify animals based on body size, habits, and habitats.
- **Artificial Method:** Proposed by Aristotle, this method was later followed by Theophrastus, Pliny, John Ray, and Linnaeus.
- **Natural System of Classification:** Later replaced the artificial method. It was based on criteria like **body organization**, **types of cells**, **chromosomes**, and **bio-chemical properties**.
- **Evolution-Based System:** Used by Dobzhansky and Meyer.
- **Recent Classification:** Proposed by Carl Woese.

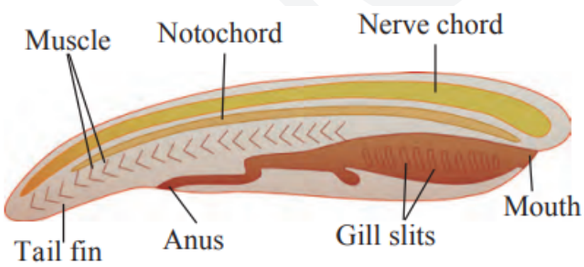
Benefits of Animal Classification

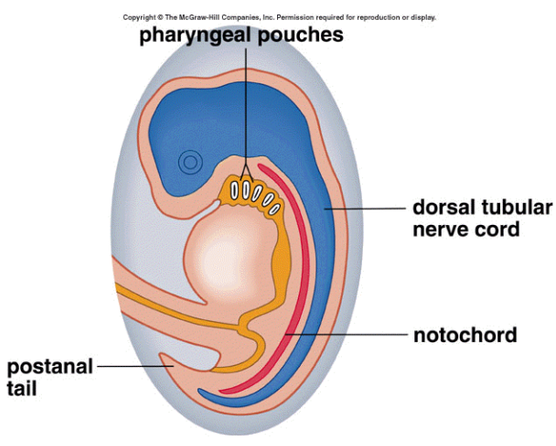
- Makes the study of animals highly convenient.
- Understanding a few animals from a specific group provides insight into the entire group.
- Provides clues about **animal evolution**.
- Allows for easy and highly accurate identification of animals.
- Clarifies the relationship between animals and other living organisms.
- Explains the habitat of each animal and its exact role in nature.
- Highlights various adaptations shown by different animals.

Traditional Method of Animal Classification

Traditionally, the **Animalia** kingdom is divided based on the presence or absence of a **notochord**.

Notochord: A long rod-like supporting structure present on the dorsal side of an animal's body. It keeps the nerve tissue isolated from the remaining body.





VS Non-Chordates vs. Chordates

Feature	❌ Non-Chordates	✅ Chordates
Notochord	The body is **not supported** by a rod-like notochord.	The body **is supported** by a notochord.
Respiration	Pharyngeal gill-slits are **absent**.	Pharyngeal gill-slits or lungs are **present**.
Nerve Cord	If present, it is on the **ventral side**, solid, and paired.	Present on the **dorsal side** and is **hollow**.
Heart	If present, it is on the **dorsal side**.	Present on the **ventral side**.

🗂️ Traditional Sub-Divisions

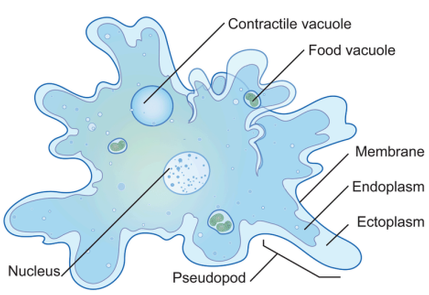
- **Non-Chordates (10 Phyla):** Protozoa, Porifera, Coelentarata/Cnidaria, Platyhelminthes, Aschelminthes, Annelida, Arthropoda, Mollusca, Echinodermata, and Hemichordata.
- **Chordates (1 Phylum):** All chordates belong to Phylum **Chordata**.
  - **Subphyla:** Urochordata, Cephalochordata, Vertebrata.
  - **Classes of Vertebrata:** Cyclostomata, Pisces, Amphibia, Reptilia, Aves, Mammalia.

🧬 Criteria for the New System of Classification

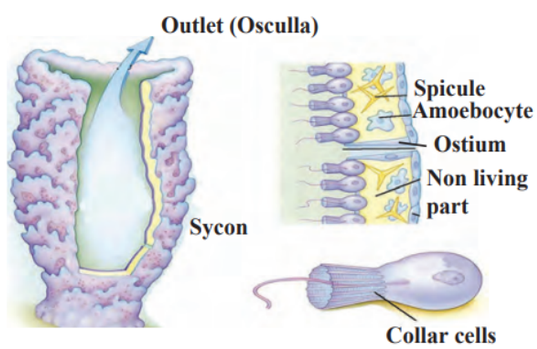
Presently, Robert Whittaker's Five-Kingdom system places all multicellular animals in **Kingdom Animalia**. The new system relies on the following structural criteria:

A. Grades of Organization

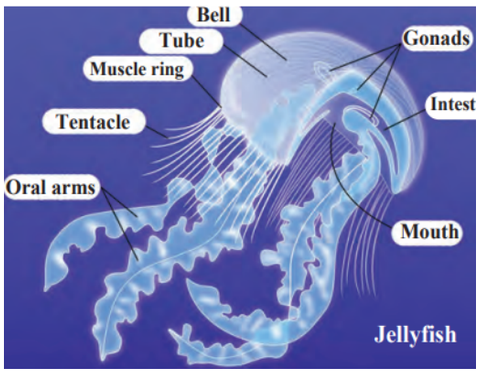
- **Protoplasmic Grade:** In unicellular organisms, all life functions are performed by a single cell.



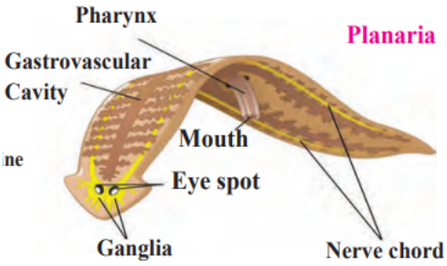
- **Cellular Grade:** Multicellular animals where cells do not form tissues (e.g., Phylum Porifera).



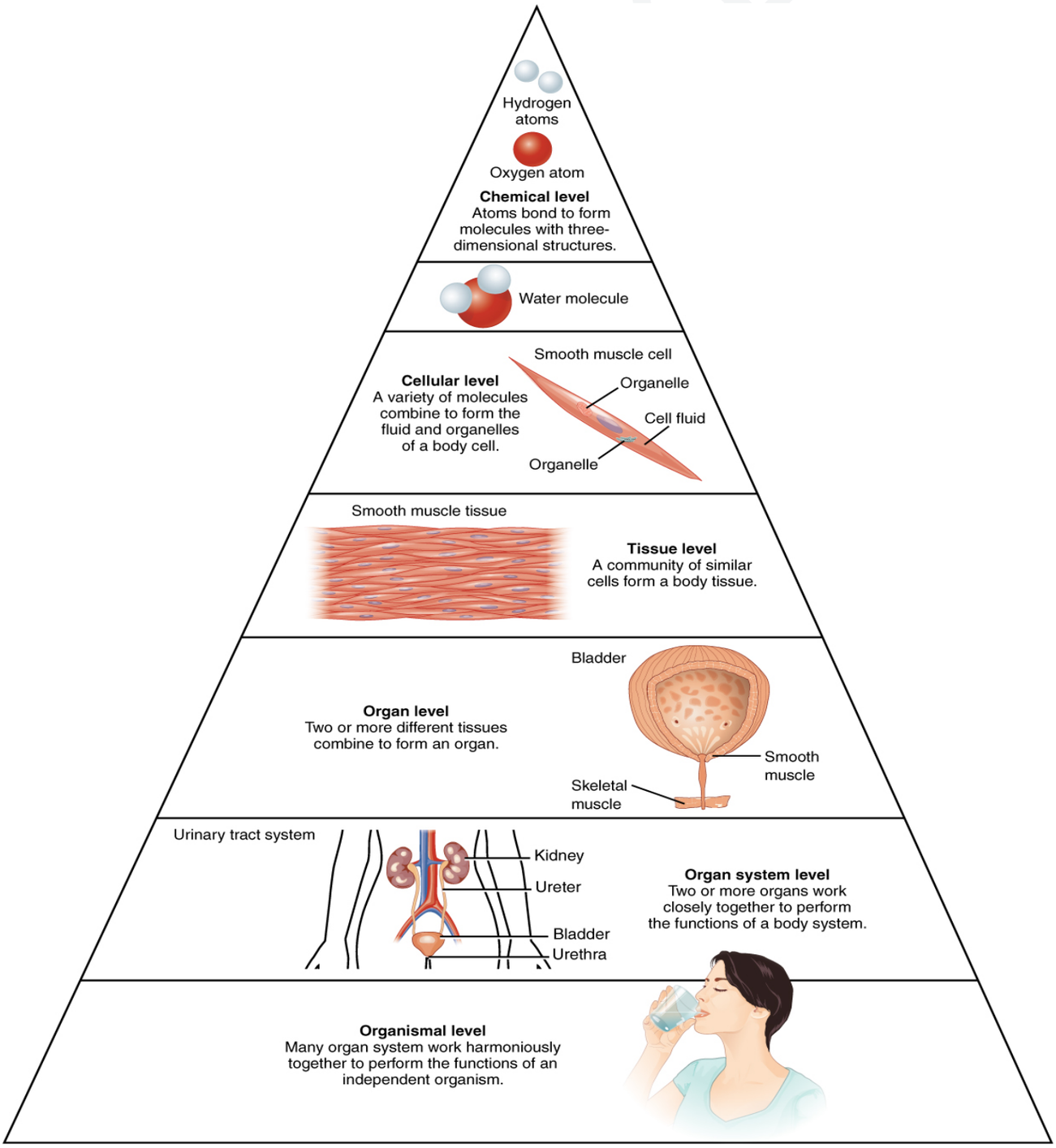
- **Cell-Tissue Grade:** Cells come together to form tissues that perform body functions (e.g., Phylum Cnidaria).



- **Tissue-Organ Grade:** Tissues organize into organs, but complete organ systems are lacking (e.g., Flatworms).



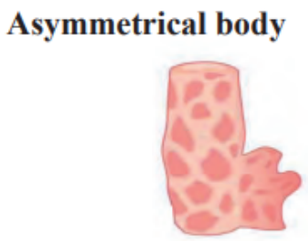
- **Organ-System Grade:** Different organs join to form specific systems (e.g., Crab, Frog, Human).



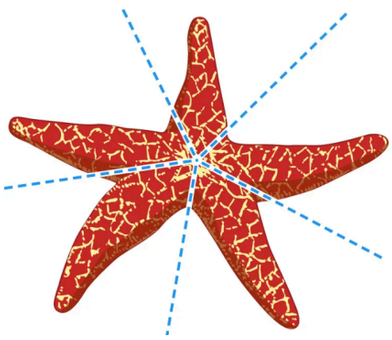
B. Body Symmetry

How a body can be divided through an imaginary central axis:

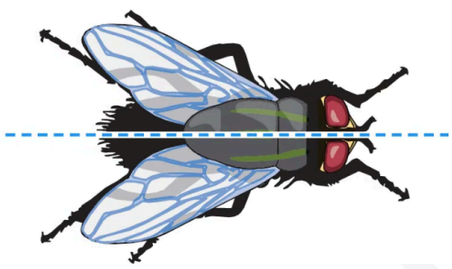
- **Asymmetrical Body:** No imaginary axis produces two equal halves (e.g., some sponges).



- **Radial Symmetry:** Any imaginary cut passing through the central axis gives two equal halves (e.g., Starfish has five such planes).



- **Bilateral Symmetry:** Only one imaginary axis divides the body into two equal halves (e.g., Insects, fishes, frog, birds, humans).



C. Germ Layers

A germ layer is a group of cells in an embryo that interact with each other as the embryo develops and contribute to the formation of all organs and tissues. All animals (except very simple ones like sponges) develop these layers through a process called gastrulation.

Layer Name	Location	Primary "Mission"	Major Organs/Tissues Formed
Ectoderm	Outer layer	To form the interface with the outside world.	Brain, spinal cord, nerves, skin (epidermis), hair, and nails.
Mesoderm	Middle layer	To form the structural and transport systems.	Muscles, bones, circulatory system (heart/blood), and reproductive system.
Endoderm	Inner layer	To form the internal linings and digestive machinery.	Lining of the digestive tract, lungs, liver, and pancreas.

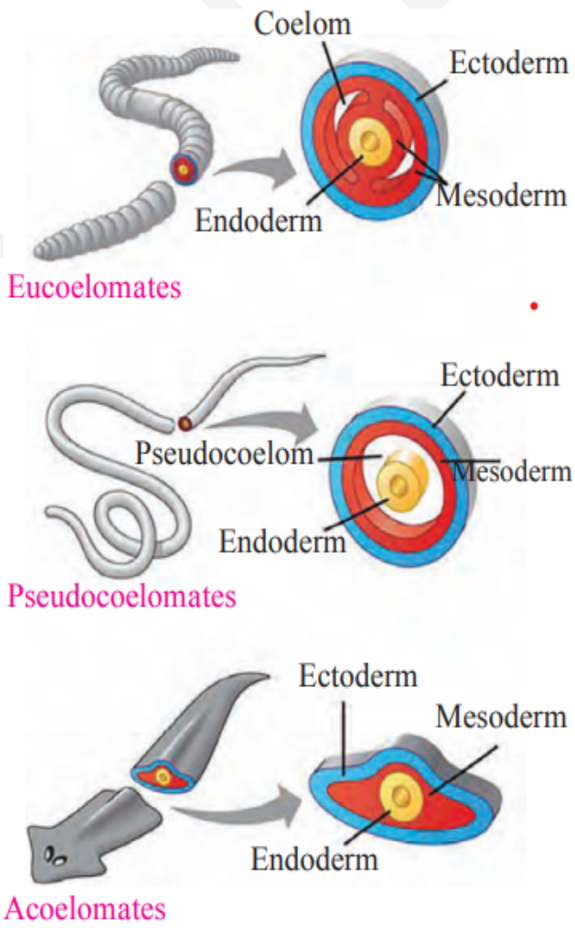
Diploblastic vs Triploblastic

Layer Type	Definition	Examples
Diploblastic	Contains only two layers: **Endoderm** & **Ectoderm**.	All Cnidarians.
Triploblastic	Contains three layers: **Endoderm**, **Ectoderm**, & **Mesoderm**.	Most remaining animals.

D. Body Cavity (Coelom)

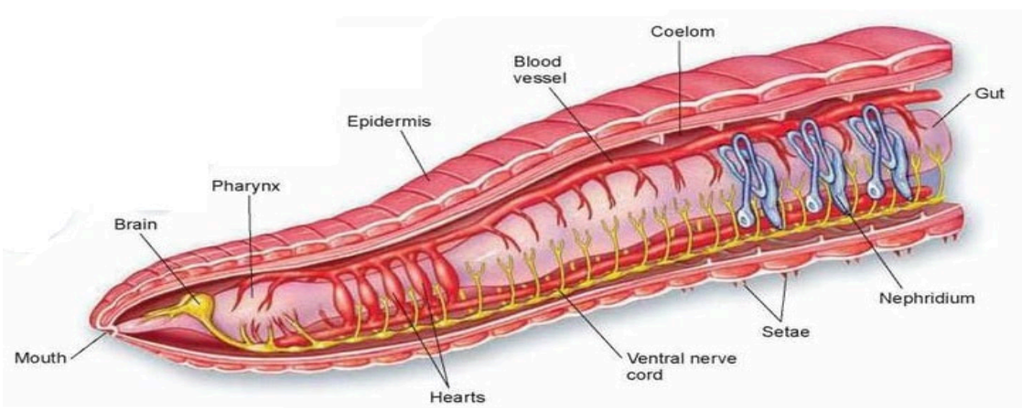
The cavity between the body wall and internal organs.

Cavity Type	Description	Phyla Examples
Eucoelomate	True body cavity formed from mesoderm during embryonic development.	Annelida and all subsequent phyla.
Acoelomate	The body cavity is entirely absent.	Porifera, Cnidaria, Platyhelminthes.
Pseudocoelomate	Body cavity exists but is not formed from the mesoderm.	Aschelminthes.



E. Body Segmentation

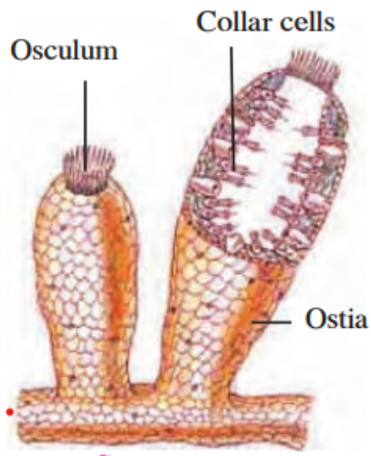
- **Segmented Body:** Animal body divided into small, similar units.
- **Segment:** Each individual small unit (e.g., Earthworm from Phylum Annelida).



Classification of Non-Chordates (Phylum by Phylum)

1. Phylum Porifera (Sponges)

- **Body Plan:** Simplest body plan; asymmetrical.
- **Pores:** Bear numerous pores called **Ostia** (water in) and **Oscula** (water out).
- **Habitat:** Aquatic (mostly marine, few freshwater).
- **Special Cells:** Possess unique **collar cells**.
- **Mobility:** Attached to a substratum (sedentary/no locomotion).
- **Skeleton:** Supported by spicules (calcium carbonate/silica) or spongin fibers.
- **Feeding:** Feed on small organisms brought in with water through the ostia.
- **Reproduction:** Asexual (budding) or sexual; excellent regeneration ability.
- **Examples:** Sycon, Euspongia (Bath sponge), Hyalonema, Euplectella.
- *Trivia:* Bath sponges hold water well due to spongin protein; historically used for bathing, pillows, and wetting currency notes.

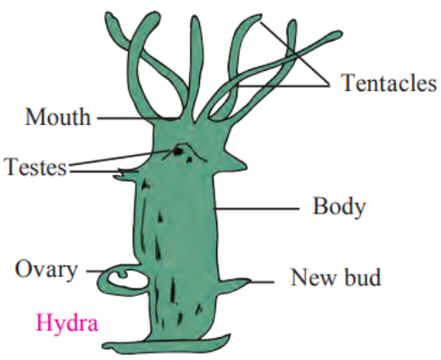


Sycon sponge

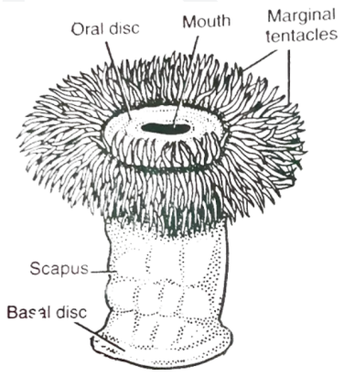
Animals in Porifera phylum

2. Phylum Coelenterata / Cnidaria

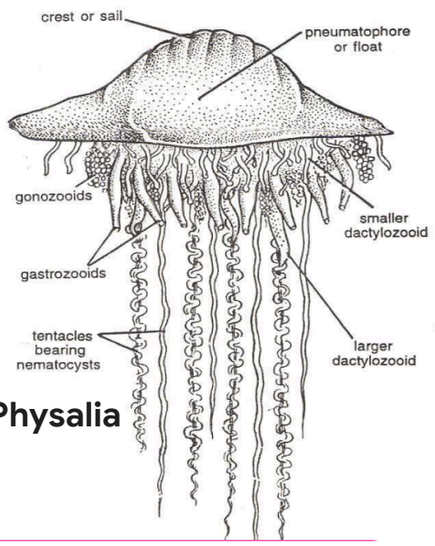
- **Body Shape:** Cylindrical (**Polyp**) or umbrella-like (**Medusa**).
- **Habitat:** Mostly marine, few freshwater.
- **Symmetry & Layers:** Radially symmetrical and diploblastic.
- **Tentacles:** Surround the mouth to capture prey.
- **Cnidoblasts:** Inject toxin into prey for hunting and protection.
- **Examples:** Hydra, Adamsia (Sea anemone), Physalia (Portuguese man-of-war), Aurelia (Jellyfish), Corals.
- *Trivia:* Coral reefs are colonies of specific cnidarians; precious stones and Ayurvedic powders are derived from them.



Hydra



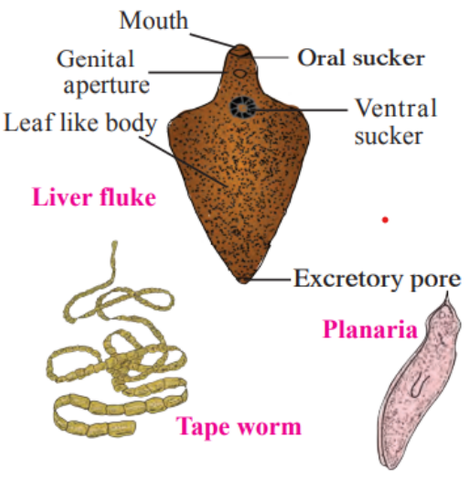
Adamsia



Physalia

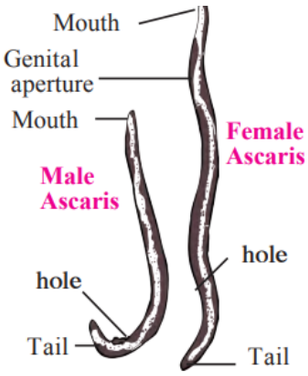
3. Phylum Platyhelminthes (Flatworms)

- **Body Shape:** Slender and flat like a leaf or strip.
- **Habitat:** Mostly endoparasites; few are free-living & aquatic.
- **Symmetry & Cavity:** Bilaterally symmetrical and acoelomate.
- **Germ Layers:** Triploblastic (endoderm, ectoderm, mesoderm).
- **Reproduction:** Hermaphrodite (male & female systems in one body).
- **Examples:** Planaria, Liver fluke, Tapeworm.



4. Phylum Aschelminthes (Roundworms)

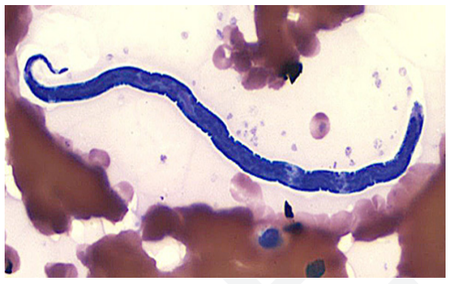
- **Body Shape:** Long, thread-like, or cylindrical.
- **Habitat:** Free-living (aquatic/terrestrial) or endoparasites.
- **Symmetry & Cavity:** Triploblastic and pseudocoelomate.
- **Protection:** Non-segmented body covered with a tough cuticle.
- **Reproduction:** Unisexual.
- **Examples:** Ascaris (Intestinal worm), Filarial worm, Loa loa (Eye worm).



Ascaris



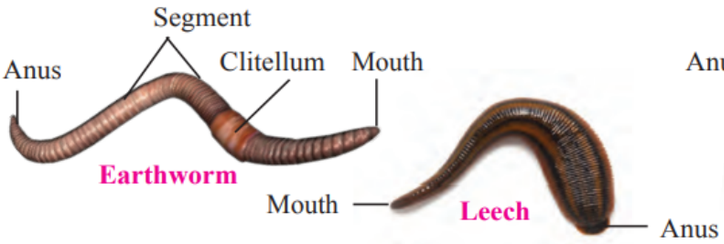
Filarial worm

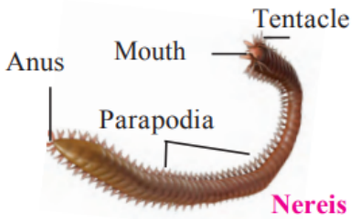


Loa loa

5. Phylum Annelida

- **Body Shape:** Long, cylindrical, and **metamerically segmented**.
- **Habitat:** Mostly free-living (marine, freshwater, terrestrial); few are ectoparasites.
- **Symmetry & Cavity:** Triploblastic, bilaterally symmetrical, and eucoelomate.
- **Locomotion:** Uses setae, parapodia, or suckers.
- **Protection:** Covered with a special cuticle.
- **Reproduction:** Hermaphrodite or unisexual.
- **Examples:** Earthworm, Leech, Nereis





6. Phylum Arthropoda

- **Defining Feature:** Possess **jointed appendages**.
- **Scale:** The **largest phylum** with the highest number of highly successful animals on Earth.
- **Habitat:** Found everywhere, from deep oceans to high mountains.
- **Symmetry & Cavity:** Triploblastic, eucoelomate, bilaterally symmetrical, and segmented.
- **Skeleton:** Covered by a **chitinous exoskeleton**.
- **Reproduction:** Unisexual.
- **Examples:** Crab, spider, scorpion, millipede, centipede, cockroach, butterfly, honey bee.



Crab



Spider



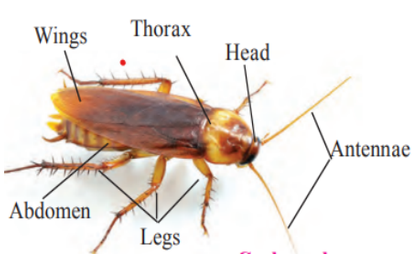
Scorpion



Centipede



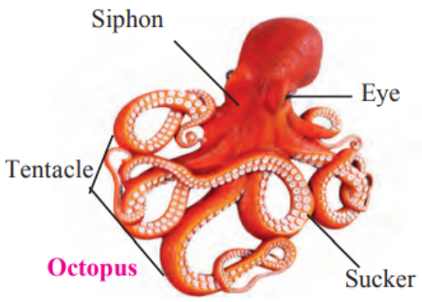
Millipede





7. Phylum Mollusca

- **Body Traits:** Soft and slimy body.
- **Scale:** The **second largest phylum** in the animal kingdom.
- **Habitat:** Aquatic (mostly marine) or terrestrial.
- **Symmetry & Cavity:** Triploblastic, eucoelomate, non-segmented, and soft. Bilateral symmetry (except snails).
- **Body Divisions:** Head, foot, and visceral mass.
- **Protection:** Visceral mass is covered with a **mantle**, which secretes a hard, calcareous shell (can be external, internal, or absent).
- **Reproduction:** Unisexual.
- **Examples:** Bivalve, Snail, Octopus.
- *Trivia:* The Octopus is the cleverest non-chordate, can change color, and performs three locomotions: swimming, creeping, and walking.




Bivalve


Snail

8. Phylum Echinodermata

- **Defining Feature:** Bear **calcareous spines** on their body.
- **Habitat:** Exclusively marine (oceans).
- **Symmetry & Cavity:** Triploblastic, eucoelomate. Radial symmetry as adults, but bilateral symmetry as larvae.
- **Locomotion/Feeding:** Use **tube-feet** to move and capture prey (some are sedentary).
- **Skeleton:** Made of calcareous spines and/or ossicles (plates).
- **Abilities:** High ability of regeneration (e.g., Starfish breaking and regrowing parts).
- **Reproduction:** Mostly unisexual.
- **Examples:** Starfish, sea-urchin, brittle star, sea-cucumber.

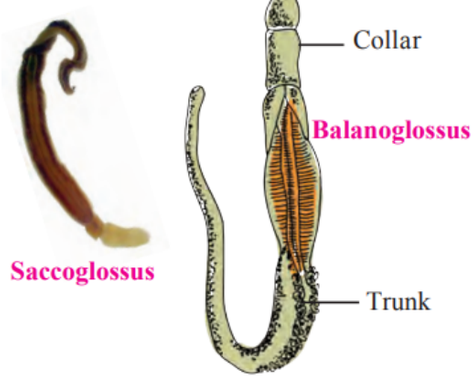

Star fish


Sea-cucumber


Sea-urchin

9. Phylum Hemichordata

- **Body Divisions:** Divided into three parts: **proboscis**, **collar**, and **trunk**.
- **Defining Feature:** Notochord is present *only* in the proboscis region.
- **Common Name:** Also known as 'acorn worms'.
- **Habitat:** Marine animals living in sand burrows.
- **Respiration:** Possess one to many pharyngeal gill slits.
- **Reproduction:** Unisexual or hermaphrodite.
- **Examples:** Balanoglossus, Saccoglossus.
- *Evolutionary Link:* Balanoglossus is considered a connecting link between non-chordates and chordates.



🦴 Phylum Chordata & Subphyla

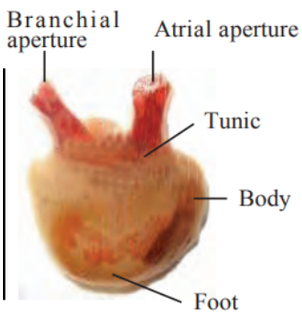
All animals with a supporting notochord belong to this single phylum.

Key Characters of Chordata:

1. Notochord present during at least one developmental stage.
2. Pharyngeal gill slits present during at least one developmental stage.
3. Single, tubular spinal cord present on the dorsal side.
4. Heart present on the ventral side.

A. Subphylum Urochordata

- **Habitat:** Marine.
- **Covering:** Body covered by a skin-like **test** or **tunic**.
- **Notochord:** Present only in the tail region of the freely swimming larvae.
- **Development:** Larvae metamorphose into adults after settling at the sea bottom.
- **Reproduction:** Generally hermaphrodite.
- **Examples:** Herdmania, Doliolum, Oikopleura.



Herdmania



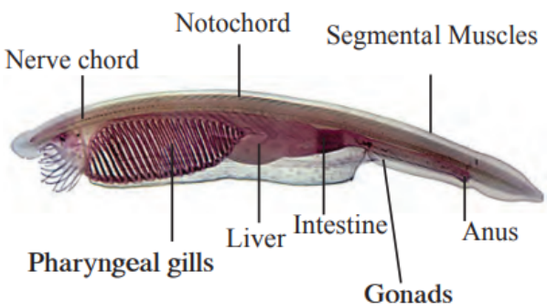
Doliolum



Oikopleura

B. Subphylum Cephalochordata

- **Body:** Small, fish-like marine animals.
- **Notochord:** Present throughout the entire body length.
- **Respiration:** Very large pharynx containing gill-slits.
- **Reproduction:** Unisexual.
- **Example:** Amphioxus.



Amphioxus

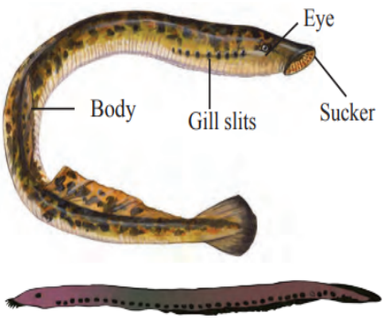
C. Subphylum Vertebrata / Craniata

- **Defining Feature:** Notochord is replaced by a **vertebral column**.
- **Structure:** Well-developed head; brain is protected by a **cranium**.
- **Skeleton:** Endoskeleton is cartilaginous or bony.
- **Jaws:** Divided into jaw-less (Agnatha) and with jaws (Gnathostomata).

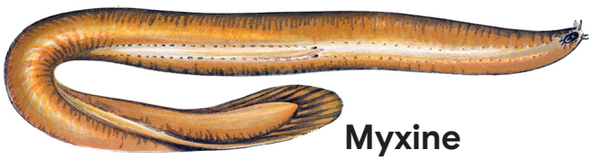
The Six Classes of Subphylum Vertebrata

a. Class Cyclostomata

- **Mouth:** Jaw-less mouth equipped with a sucker.
- **Skin:** Soft and devoid of scales.
- **Appendages:** Paired appendages are absent.
- **Skeleton:** Cartilaginous endoskeleton.
- **Lifestyle:** Mostly ectoparasites.
- **Examples:** Petromyzon, Myxine.



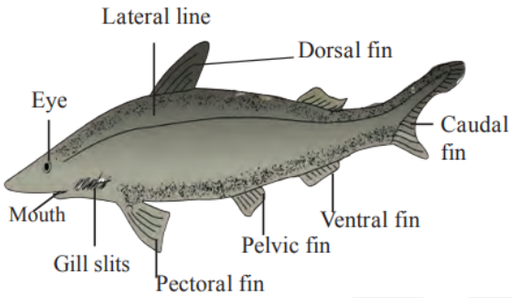
Petromyzon



Myxine

b. Class Pisces

- **Blood Type:** Cold-blooded (Poikilotherms).
- **Habitat:** Aquatic (marine and fresh water).
- **Body Shape:** Spindle-shaped to minimize water resistance.
- **Lococmotion:** Paired & unpaired fins for swimming; tail fin acts as a steering organ.
- **Skeleton:** Exoskeleton has scales; endoskeleton is cartilaginous or bony.
- **Respiration:** Via gills.
- **Examples:** Rohu, Pomfret, Sea horse, Shark, Electric ray, Sting ray



Scoliodon



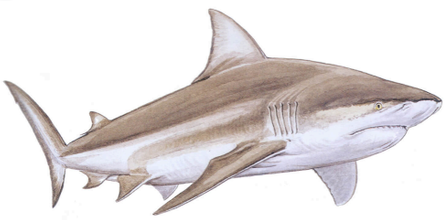
Rohu



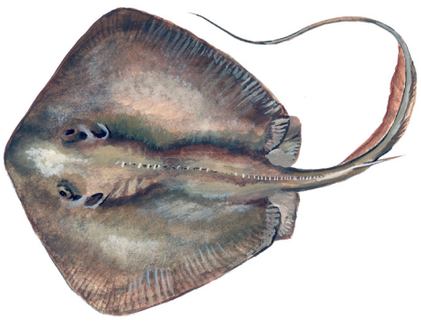
Pomfret



Sea horse



Shark

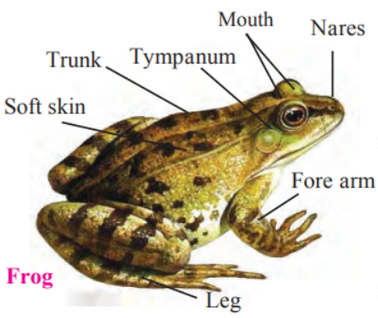


Sting ray

c. Class Amphibia

- **Life Cycle:** Strictly aquatic in larval life (aquatic respiration); land and water dwellers in adult life (aquatic and aerial respiration).
- **Appendages:** Two pairs of appendages; digits lack claws.
- **Skin:** No exoskeleton; skin is mostly moist for respiration.

- **Senses:** External ear is absent, but a **tympanum** is present. Prominent eyes with eyelids.
- **Structure:** Neck is absent.
- **Examples:** Frog, Toad, Salamander.



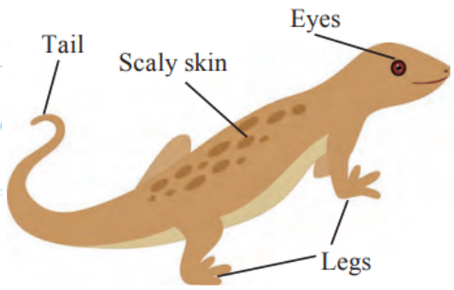




Frog **Toad** **Salamander**

d. Class Reptilia

- **Evolution:** The first true terrestrial animals with creeping movement.
- **Blood Type:** Cold-blooded (poikilotherms).
- **Movement:** Creep on land (body cannot be lifted up).
- **Skin:** Dry and scaly.
- **Structure:** Neck is present between head and trunk. External ear is absent.
- **Appendages:** Digits are provided with claws.
- **Examples:** Tortoise, Lizard, Snake.



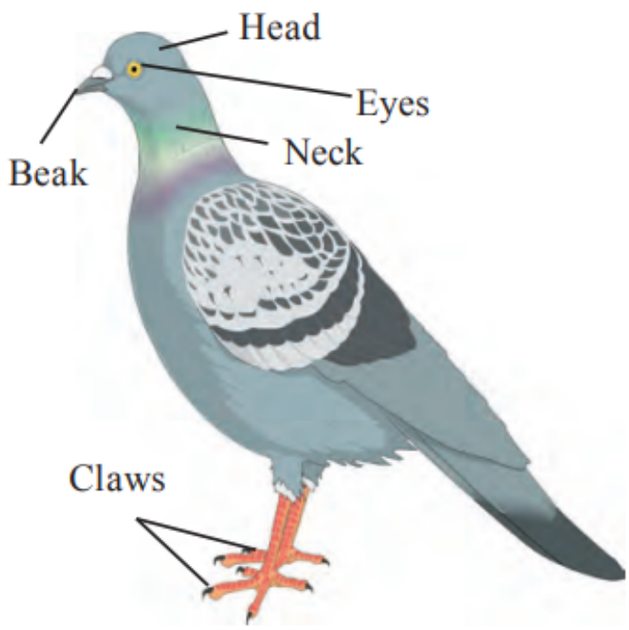




Lizard **Tortoise** **Snake**

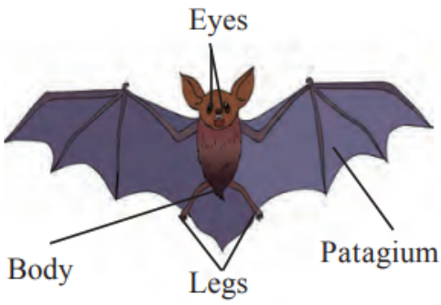
e. Class Aves

- **Adaptation:** Completely adapted for aerial life.
- **Blood Type:** Warm-blooded (Homeotherms - maintain constant body temperature).
- **Body Shape:** Spindle-shaped to minimize air resistance.
- **Appendages:** Forelimbs modified into wings. Digits have scales and claws.
- **Covering:** Exoskeleton consists of feathers.
- **Structure:** Neck present between head and trunk. Jaws modified into a beak.
- **Examples:** Peacock, Parrot, Pigeon, Duck, Penguin.



f. Class Mammalia 🐘

- **Defining Feature:** Presence of **mammary glands**.
- **Blood Type:** Warm-blooded.
- **Body Divisions:** Head, neck, trunk, and tail.
- **Appendages:** Digits equipped with nails, claws, or hooves.
- **Covering:** Exoskeleton is in the form of hairs or fur.
- **Examples:** Elephant, Human, Kangaroo, Dolphin, Bat.



🧠 Textbook Reflection Questions to Ponder:

- *Why is the earthworm called the friend of farmers?*
- *How is the leech used in Ayurvedic treatments?*
- *What are the benefits and harms caused by Arthropods?*
- *Are ghariyals and crocodiles amphibians or reptiles? * Are whales and walruses included in Pisces or Mammalia? ---*

🎯 Key Takeaways

- **Classification** allows scientists to organize and understand the estimated 7 million animal species effectively.
- The system has evolved from Aristotle's Artificial method to the highly precise **Five-Kingdom System** driven by body organization, germ layers, symmetry, and coelom presence.
- **Non-chordates** lack a true notochord and are broken into 10 distinct phyla ranging from simple sponges (Porifera) to complex Echinoderms.
- **Chordates** possess a notochord at some life stage and feature a highly diverse subphylum, Vertebrata, split into six classes including humans (Mammalia), birds (Aves), and fishes (Pisces).
- Always remember to study and observe the varieties of animals in our surroundings with caution and care, ensuring no harm comes to them!

LUCKY YELE